EXP 3 B

Aim

To optimize the directional coupler length by 5% and compare its coupling response before and after optimization.

Software required:

MATLAB

Program:

```
clc;
clear;
close all;

% Part B: Length Optimization
f0 = 10e9;
f = linspace(8e9,12e9,401);
c = 3e8;

% Initial and optimized lengths
lambda0 = c/f0;
L_initial = lambda0/4;
L_optimized = 1.05*L_initial;

% Coupling coefficient for 10 dB
coupling
k = 10^(-10/20);

% Propagation constant
beta = 2*pi*f/c;

% Initial design response
theta_initial = beta*L_initial;
S21_initial = cos(theta_initial);
S31_initial =
1i*k.*sin(theta_initial);

% Optimized design response
theta_optimized =
beta*L_optimized;
S21_optimized =
cos(theta_optimized);
S31_optimized =
1i*k.*sin(theta_optimized);

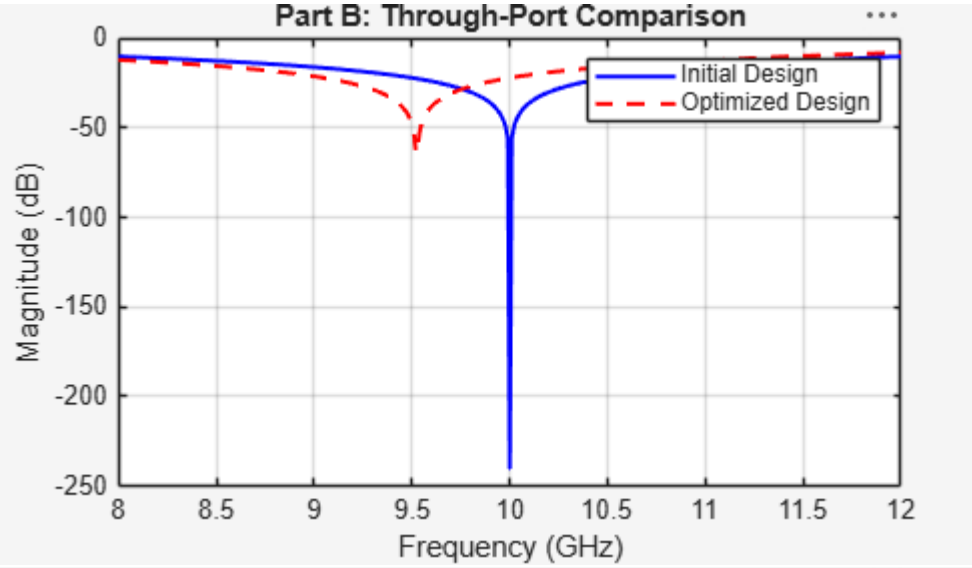
% Convert to dB
S21_initial_dB =
20*log10(abs(S21_initial)+1e-12);
S31_initial_dB =
20*log10(abs(S31_initial)+1e-12);
S21_optimized_dB =
20*log10(abs(S21_optimized)+1e-12);
S31_optimized_dB =
20*log10(abs(S31_optimized)+1e-12);

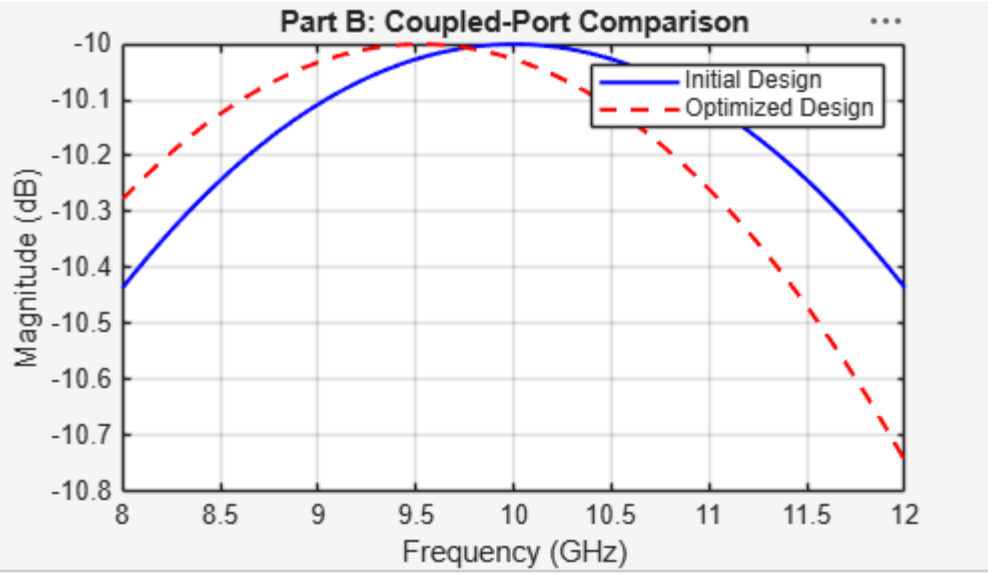
% Compare through response plots
figure;
plot(f/1e9,S21_initial_dB,'b','LineWidth',1.5);
hold on;
plot(f/1e9,S21_optimized_dB,'r--
','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Part B: Through-Port Comparison');
legend('Initial Design','Optimized Design');

figure;
plot(f/1e9,S31_initial_dB,'b','LineWidth',1.5);
hold on;
plot(f/1e9,S31_optimized_dB,'r--
','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Part B: Coupled-Port Comparison');
legend('Initial Design','Optimized Design');

fprintf('Part B: Optimization Results\n');
fprintf('Initial length = %.3f
mm\n',L_initial*1e3);
fprintf('Optimized length = %.3f
mm\n',L_optimized*1e3);
fprintf('Length increase = %.1f percent\n', ...
    100*(L_optimized-L_initial)/L_initial);
```


Output:






```
Part B: Optimization Results
Initial length = 7.500 mm
Optimized length = 7.875 mm
Length increase = 5.0 percent
```